

Task 3b

Evaluation of the Developed Solution

Greenfield Local Hub

Introduction

This document evaluates the Greenfield Local Hub website — a Flask-based local produce e-commerce platform connecting customers with local farmers and independent producers. The evaluation covers the assets and content used during development, how well the functional and non-functional requirements were met, and uses evidence from the Task 3a feedback process to support all claims. The document concludes with realistic, evidence-backed improvements.

A. Assets & Content

Key Assets Chosen and Why

Flask (Python Web Framework) was chosen as the core framework because it is lightweight and flexible, giving full control over the application structure without imposing unnecessary complexity. It provided all the tools needed — routing, session management, Jinja2 templating, and flash messaging — to build the platform's features including user authentication, the shopping cart, checkout, and role-based access for customers and producers. Django was considered as an alternative but rejected because its built-in ORM and admin interface would have been excessive for a focused, single-developer project of this scale.

SQLite was selected as the database because it is serverless, requires no additional configuration, and stores the entire database in a single file. This made it ideal for a development and demonstration context. PostgreSQL was considered as an alternative as it offers better performance and concurrency for production environments, but was rejected because it requires a running server process and additional configuration that would add unnecessary complexity at this stage.

Custom CSS was written from scratch rather than using a framework like Bootstrap. This gave full control over the visual design and avoided large external dependencies. The dark green colour scheme was chosen to reflect the fresh, natural identity of a local produce marketplace. Bootstrap was rejected because its default styles would have required significant overriding to achieve the desired look.

Legal and Ethical Issues

The product images used on the platform are hosted on third-party websites and were not explicitly licensed for reuse. In a real-world deployment this could represent a copyright issue, as the intellectual property rights belong to the original creators. Future versions should use images sourced from royalty-free libraries such as Unsplash or Pexels, or self-produced photography. The Flask and SQLite libraries used are open-source and licensed under the BSD licence and public domain respectively, meaning they can be used freely without restriction.

From an ethical perspective, the platform was designed to be inclusive and accessible to users of all technical abilities, with a consistent navigation structure and clear flash messaging to guide users through each step. However, as highlighted by one non-technical respondent who requested “more accessibility features,” the platform currently lacks ARIA labels, formal colour contrast testing against WCAG standards, and adjustable font sizes — all of which are important for users with visual impairments or disabilities and should be addressed in future development.

B. Requirements Evaluation

Functional Requirements

User Authentication was fully met. The registration system validates all fields, enforces a minimum password length, hashes passwords using SHA-256, and uses Flask’s session system to maintain login state. Role-based access correctly separates customer and producer journeys. All five technical respondents confirmed the system meets its intended purpose and that authentication is secure.

Product Browsing and Search was fully met. The Browse Products page renders products dynamically from the SQLite database with filtering by category and keyword search. All ten non-technical respondents were able to find the Browse Products page without difficulty, and the product grid was rated as organised by the majority of users, scoring predominantly 4 or 5 out of 5. However, one respondent rated search result relevance at only 1 out of 5, suggesting the basic LIKE query may not meet the expectations of more experienced online shoppers.

Cart and Checkout was fully met. Users can add, update, and remove items, and the checkout supports both collection and delivery options with dynamic JavaScript toggling of the address form. Stock is validated before orders are placed. Non-technical feedback for the cart and checkout was consistently positive, with the majority of respondents rating confidence in placing an order at 4 or 5 out of 5.

Loyalty Rewards System was technically met but partially failed on the user-facing side. The backend correctly awards 10 points per £1 spent and automatically upgrades user tiers. Technical respondents rated the loyalty system between 4 and 5 out of 5. However, non-technical feedback showed that some users did not fully understand how the system works, indicating the front-end communication of the feature falls short of what is needed for users to fully benefit from it.

Non-Functional Requirements

Performance was fully met. Two technical respondents explicitly praised the load times, with one describing performance as “optimal with near-instantaneous loading times.” No non-technical users reported performance as a concern.

Usability was largely met. Overall ease of use was rated 4 or 5 out of 5 by the majority of non-technical respondents, and no users reported feeling lost during navigation. The consistent navbar across all pages and clear flash messaging contributed to an intuitive experience.

Responsiveness was partially met. Two technical respondents rated responsiveness at only 3 out of 5, reflecting that while the CSS uses flexbox and grid layouts, no formal `@media` query breakpoints are defined in `style.css`. Mobile users could complete key journeys but the experience is not fully optimised for smaller screens.

Security was partially met. Passwords are hashed and routes are protected by decorators, meeting basic security requirements. However, the AI respondent recommended “More Security” as their primary suggestion, and a code review confirms that SHA-256 is used without salting and the Flask secret key is hardcoded — both of which fall below best practice standards for a production environment.

Performance Against KPIs

The majority of KPIs were met. Navigation clarity was fully achieved — all ten non-technical users found the Browse Products page without assistance. Overall user satisfaction exceeded the target, with most respondents rating ease of use at 4 or 5 out of 5. Technical quality met the target, with all five technical respondents rating the system 4 or 5 out of 5. Feature completeness was largely achieved, though minor bugs including the “None” field rendering and broken image URLs were identified. Security was only partially met,

representing the most significant gap against the KPIs.

C. Use of Evidence (Task 3a Link)

The evaluations made throughout this document are directly supported by evidence from Task 3a. The claim that navigation is effective is triangulated across quantitative scores — seven of ten non-technical respondents rated navigation ease at 4 or 5 out of 5 — the observational theme that no users felt lost, and the fact that all respondents confirmed the menu options were clear. This combination of score, theme, and observation provides strong evidence that the navigation requirement was fully met.

The claim that visual design requires improvement is supported by three independent user quotes collected in Task 3a. Respondent 3 stated that “higher detailed UI design” was needed as “slightly more basic looking websites can plant seeds of doubt within customers.” Respondent 8 commented that “a more visually pleasing website design would significantly improve the overall feel of the website and entice customers to browse.” Respondent 6 suggested the platform should be made “more engaging.” These quotes are corroborated by the cybersecurity technical respondent who noted that “the design is a bit simple.” The convergence of non-technical user opinion and technical peer observation through triangulation provides the strongest evidence base of any finding in this evaluation.

The claim regarding the loyalty system’s communication gap is supported by triangulating the high technical ratings (4–5 out of 5) against the inconsistent comprehension scores from non-technical users, some of whom indicated they did not fully understand how the points system works. This clearly locates the issue in the front-end presentation rather than the backend implementation, which is a precise and useful insight for future development.

D. Improvements

1. Fix the “None” Field Rendering in Form Fields

The most immediately actionable improvement is resolving the display of the word “None” in address, city, and phone number fields on the checkout and customer dashboard pages. This occurs because Jinja2 renders Python’s `None` database value as the string “None” when a user’s profile details have not been saved. The fix is straightforward — updating templates to use `{{ user.address or '' }}` syntax will display blank fields instead. This improvement is directly supported by Respondent 7’s Task 3a feedback, who specifically stated that “instead of having the text ‘None’ on the address, city, phone number put in nothing.” This matters because displaying “None” in form fields is unprofessional and

erodes the user trust that is essential for an e-commerce platform asking users to enter personal and delivery information.

2. Improve Visual Design and UI Engagement

The visual design of Greenfield Local Hub should be enhanced to create a more polished and engaging user experience. This could include richer homepage imagery, improved card styling on the product grid, and stronger typographic hierarchy. This improvement is supported by the consistent theme across Task 3a feedback — three non-technical respondents independently raised the basic appearance of the website in their open-ended suggestions, and the cybersecurity technical respondent echoed this view. As Respondent 3 noted, a basic-looking website can cause customers to doubt its legitimacy, which directly impacts the platform's ability to convert browsers into buyers. Improving the visual design would build greater customer trust and better represent the quality of the local produce sold on the platform.

3. Upgrade Security Implementation

The password hashing implementation should be upgraded from `hashlib.sha256` to `werkzeug.security's generate_password_hash` function, which uses PBKDF2 with a unique salt per password. The Flask `secret_key` should also be moved from the hardcoded string in `app.py` to an environment variable. This improvement is supported by the AI respondent's primary suggestion of "More Security" and the cybersecurity respondent's identification of potential vulnerabilities. SHA-256 without salting is vulnerable to rainbow table attacks, meaning that if the database were ever compromised, common passwords could be recovered quickly by an attacker. These improvements would protect all users' personal data and order histories, which is particularly important given that the platform stores delivery addresses and purchase histories.

4. Add Formal Mobile Responsiveness

CSS `@media` query breakpoints should be added to `style.css` to provide a fully optimised layout for mobile and tablet screen sizes. Key changes would include adapting the four-column product grid to a single column on mobile and implementing a responsive navbar. This improvement is supported by two technical respondents rating responsiveness at only 3 out of 5 — the lowest score given for any layout-related metric in the technical feedback. Mobile optimisation is essential for Greenfield Local Hub because its target audience of everyday grocery shoppers are highly likely to browse and purchase on their smartphones. A fully responsive layout would remove friction from the mobile purchasing journey and make the platform accessible to a significantly wider audience.

Conclusion

Overall, Greenfield Local Hub successfully meets the majority of its functional and non-functional requirements, delivering a complete e-commerce user journey from product browsing through to order placement and loyalty rewards. The Task 3a feedback confirms that the platform is intuitive and well-structured, with non-technical users rating overall ease of use at 4 or 5 out of 5 and technical users rating the system equally highly. However, the evaluation has identified clear gaps in visual design, mobile responsiveness, security implementation, and front-end communication of the loyalty system. The four improvements proposed are each directly grounded in Task 3a evidence and are realistic within the scope of the project. Addressing these improvements would elevate Greenfield Local Hub from a strong working prototype into a polished, secure, and fully accessible platform that genuinely serves the needs of local producers and their customers.